Assignment_4

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mysql> CREATE DATABASE CompanyDB;
Query OK, 1 row affected (0.02 sec)

mysql> ^C
mysql> USE CompanyDB;
Database changed
mysql> CREATE TABLE Department (
    ->     DeptID INT PRIMARY KEY,
    ->     DeptName VARCHAR(50) NOT NULL,
    ->     Location VARCHAR(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO Department (DeptID, DeptName, Location) VALUES
    -> (1, 'HR', 'Mumbai'),
    -> (2, 'IT', 'Pune'),
    -> (3, 'Finance', 'Mumbai'),
    -> (4, 'Marketing', 'Delhi'),
    -> (5, 'Sales', 'Nagpur');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Emp (
    ->     EmpID INT AUTO_INCREMENT PRIMARY KEY,
    ->     EmpName VARCHAR(50) NOT NULL,
    ->     Salary DECIMAL(10, 2) NOT NULL,
    ->     DeptID INT,
    ->     FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Emp (EmpName, Salary, DeptID) VALUES
    -> ('Amit Sharma', 45000.00, 1),
    -> ('Priya Verma', 65000.00, 2),
    -> ('Rahul Patil', 55000.00, 3),
    -> ('Sneha Joshi', 40000.00, NULL),
    -> ('Vikram Singh', 75000.00, 2);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Project (
    ->     ProjectID INT PRIMARY KEY,
    ->     ProjectName VARCHAR(50) NOT NULL,
    ->     Client VARCHAR(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Project (ProjectID, ProjectName, Client) VALUES
    -> (101, 'Alpha CRM', 'TechCorp'),
    -> (102, 'Beta ERP', 'GlobalTech'),
    -> (103, 'Gamma Portal', 'RetailPro'),
    -> (104, 'Delta Analytics', 'DataCo'),
    -> (105, 'Omega Cloud', 'CloudNet');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT e.EmpName, d.DeptName
    -> FROM Emp e
    -> INNER JOIN Department d ON e.DeptID = d.DeptID;
+--------------+----------+
| EmpName      | DeptName |
+--------------+----------+
| Amit Sharma  | HR       |
| Priya Verma  | IT       |
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| Rahul Patil  | Finance  |
| Vikram Singh | IT       |
+--------------+----------+
4 rows in set (0.00 sec)

mysql> SELECT e.EmpID, e.EmpName, e.Salary, d.DeptName, d.Location
    -> FROM Emp e
    -> LEFT OUTER JOIN Department d ON e.DeptID = d.DeptID;
+-------+--------------+----------+----------+----------+
| EmpID | EmpName      | Salary   | DeptName | Location |
+-------+--------------+----------+----------+----------+
|     1 | Amit Sharma  | 45000.00 | HR       | Mumbai   |
|     2 | Priya Verma  | 65000.00 | IT       | Pune     |
|     3 | Rahul Patil  | 55000.00 | Finance  | Mumbai   |
|     4 | Sneha Joshi  | 40000.00 | NULL     | NULL     |
|     5 | Vikram Singh | 75000.00 | IT       | Pune     |
+-------+--------------+----------+----------+----------+
5 rows in set (0.00 sec)

mysql> SELECT e.EmpID, e.EmpName, d.DeptID, d.DeptName, d.Location
    -> FROM Emp e
    -> RIGHT OUTER JOIN Department d ON e.DeptID = d.DeptID;
+-------+--------------+--------+-----------+----------+
| EmpID | EmpName      | DeptID | DeptName  | Location |
+-------+--------------+--------+-----------+----------+
|     1 | Amit Sharma  |      1 | HR        | Mumbai   |
|     2 | Priya Verma  |      2 | IT        | Pune     |
|     5 | Vikram Singh |      2 | IT        | Pune     |
|     3 | Rahul Patil  |      3 | Finance   | Mumbai   |
|  NULL | NULL         |      4 | Marketing | Delhi    |
|  NULL | NULL         |      5 | Sales     | Nagpur   |
+-------+--------------+--------+-----------+----------+
6 rows in set (0.00 sec)

mysql> SELECT *
    -> FROM Emp
    -> NATURAL JOIN Department;
+--------+-------+--------------+----------+----------+----------+
| DeptID | EmpID | EmpName      | Salary   | DeptName | Location |
+--------+-------+--------------+----------+----------+----------+
|      1 |     1 | Amit Sharma  | 45000.00 | HR       | Mumbai   |
|      2 |     2 | Priya Verma  | 65000.00 | IT       | Pune     |
|      3 |     3 | Rahul Patil  | 55000.00 | Finance  | Mumbai   |
|      2 |     5 | Vikram Singh | 75000.00 | IT       | Pune     |
+--------+-------+--------------+----------+----------+----------+
4 rows in set (0.00 sec)

mysql> SELECT e.EmpID, e.EmpName, p.ProjectID, p.ProjectName
    -> FROM Emp e
    -> CROSS JOIN Project p;
+-------+--------------+-----------+-----------------+
| EmpID | EmpName      | ProjectID | ProjectName     |
+-------+--------------+-----------+-----------------+
|     5 | Vikram Singh |       101 | Alpha CRM       |
|     4 | Sneha Joshi  |       101 | Alpha CRM       |
|     3 | Rahul Patil  |       101 | Alpha CRM       |
|     2 | Priya Verma  |       101 | Alpha CRM       |
|     1 | Amit Sharma  |       101 | Alpha CRM       |
|     5 | Vikram Singh |       102 | Beta ERP        |
|     4 | Sneha Joshi  |       102 | Beta ERP        |
|     3 | Rahul Patil  |       102 | Beta ERP        |
|     2 | Priya Verma  |       102 | Beta ERP        |
|     1 | Amit Sharma  |       102 | Beta ERP        |
|     5 | Vikram Singh |       103 | Gamma Portal    |
|     4 | Sneha Joshi  |       103 | Gamma Portal    |
|     3 | Rahul Patil  |       103 | Gamma Portal    |
|     2 | Priya Verma  |       103 | Gamma Portal    |
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|     1 | Amit Sharma  |       103 | Gamma Portal    |
|     5 | Vikram Singh |       104 | Delta Analytics |
|     4 | Sneha Joshi  |       104 | Delta Analytics |
|     3 | Rahul Patil  |       104 | Delta Analytics |
|     2 | Priya Verma  |       104 | Delta Analytics |
|     1 | Amit Sharma  |       104 | Delta Analytics |
|     5 | Vikram Singh |       105 | Omega Cloud     |
|     4 | Sneha Joshi  |       105 | Omega Cloud     |
|     3 | Rahul Patil  |       105 | Omega Cloud     |
|     2 | Priya Verma  |       105 | Omega Cloud     |
|     1 | Amit Sharma  |       105 | Omega Cloud     |
+-------+--------------+-----------+-----------------+
25 rows in set (0.00 sec)

mysql> CREATE TABLE EmpBackup AS
    -> SELECT * FROM Emp
    -> WHERE 1 = 0;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> INSERT INTO EmpBackup
    -> SELECT * FROM Emp;
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE ProjectArchive AS
    -> SELECT * FROM Project;
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Emp (
    ->     EmpID INT AUTO_INCREMENT PRIMARY KEY,
    ->     EmpName VARCHAR(50) NOT NULL,
    ->     Salary DECIMAL(10, 2) NOT NULL,
    ->     DeptID INT
    -> );
ERROR 1050 (42S01): Table 'emp' already exists
mysql> SELECT EmpID, EmpName, Salary
    -> FROM Emp
    -> WHERE Salary > (SELECT AVG(Salary) FROM Emp);
+-------+--------------+----------+
| EmpID | EmpName      | Salary   |
+-------+--------------+----------+
|     2 | Priya Verma  | 65000.00 |
|     5 | Vikram Singh | 75000.00 |
+-------+--------------+----------+
2 rows in set (0.00 sec)

mysql> SELECT EmpID, EmpName, Salary, DeptID
    -> FROM Emp
    -> WHERE DeptID IN (SELECT DeptID FROM Department WHERE Location = 'Mumbai');
+-------+-------------+----------+--------+
| EmpID | EmpName     | Salary   | DeptID |
+-------+-------------+----------+--------+
|     1 | Amit Sharma | 45000.00 |      1 |
|     3 | Rahul Patil | 55000.00 |      3 |
+-------+-------------+----------+--------+
2 rows in set (0.00 sec)


mysql> SELECT EmpID, EmpName, Salary
    -> FROM Emp
    -> WHERE Salary = (SELECT MAX(Salary) FROM Emp);
+-------+--------------+----------+
| EmpID | EmpName      | Salary   |
+-------+--------------+----------+
|     5 | Vikram Singh | 75000.00 |
```

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+-------+--------------+----------+
1 row in set (0.00 sec)

mysql> SELECT e1.EmpName AS Employee1, e2.EmpName AS Employee2, e1.DeptID
    -> FROM Emp e1
    -> JOIN Emp e2 ON e1.DeptID = e2.DeptID AND e1.EmpID < e2.EmpID;
+-------------+--------------+--------+
| Employee1   | Employee2    | DeptID |
+-------------+--------------+--------+
| Priya Verma | Vikram Singh |      2 |
+-------------+--------------+--------+
1 row in set (0.00 sec)

mysql> SELECT d1.DeptName AS Dept1, d2.DeptName AS Dept2, d1.Location
    -> FROM Department d1
    -> JOIN Department d2 ON d1.Location = d2.Location AND d1.DeptID < d2.DeptID;
+-------+---------+----------+
| Dept1 | Dept2   | Location |
+-------+---------+----------+
| HR    | Finance | Mumbai   |
+-------+---------+----------+
1 row in set (0.00 sec)
```